

Results summary

1 Method of detection

1. **Fault model** instruction skip: step the PC past a `bl`, so that one operation never executes. Applied to all 20 operation-calls in the signing loop, one at a time.
2. **Feature the matched filter**: correlation of the sparse challenge c against each response polynomial z (a 1024-vector, $L = 4$ polys $\times$ 256 lags). The *marginal* of z is secret-independent by design, so a plain histogram of z is blind; the leak lives only in the joint (c, z) .
3. **Verdict, two equivalent ways**:
 - **Two-key classifier accuracy** (leave-one-out). Signed score $\text{score}(x) = \langle x, g_{\text{own}}^{\text{LOO}} \rangle - \langle x, g_{\text{other}} \rangle$ against per-key mean templates; key-A sigs should score > 0 , key-B < 0 . 50% = indistinguishable, 100% = leak, flag at $\geq 80\%$.
 - **TVLA t-test** (`ttest_09.py`). Welch two-sample t at every one of the 1024 matched-filter coordinates between key A and key B; flag any coordinate with $|t| > 4.5$ ($\approx p < 10^{-5}$).

The two verdicts agree on every site.

2 Positive result

site	addr	two-key acc	TVLA max $ t $	#coords > 4.5	perm p
<code>z = z + y (polyvecl_add)</code>	<code>0x4d5e</code>	100%	15.4	444 / 1024	0.0005 (floor)

3 Negative results

Two distinct kinds of non-leak.

3.1 (a) Runs fine, but no leak

Nine sites complete signing yet stay at chance. The mask survives (or the skipped step doesn't touch it), so key A and key B overlap on the decision boundary.

site	addr	acc (N=16)	acc (N=28)	TVLA max $ t $	#coords > 4.5
control (no fault)	—	53%	52%	3.6	0
memcpy	0x4c90	47%	46%	3.6	0
$y \rightarrow \text{NTT}$	0x4c9a	41%	50%	3.6	0
$w = A \cdot y$	0x4cb0	41%	54%	3.4	0
$w \rightarrow \text{invNTT}$	0x4cc4	75%	46%	3.9	0
pack w_1	0x4cec	69%	50%	3.7	0
shake squeeze $\tilde{c}$	0x4d18	47%	(hang at $N=28$)	—	—
reduce z	0x4d68	38%	46%	3.5	0
pack z into sig	0x4d88	41%	59%	3.3	0
$w_0 - c \cdot s_2$	0x4dca	59%	59%	3.6	0

3.2 (b) Hangs because the skip breaks the loop

Since Signing is reject-and-retry; a whole-call skip drops a value a rejection check depends on, the check fails every iteration, and the loop never accepts.

```
sample y · decompose w · poly_challenge · c*s1 · z-norm check · c*s2 · r0 check ·
c*t0 · ct0 check · make_hint
```

The important negative: the three **rejection-check** skips (**z-norm**, **r0**, **ct0**) hang rather than leak, even though those checks *are* the zero-knowledge mask and faulting them **does** leak. The reason is that a whole-call skip breaks the loop because the next check sees a stale value, and the loop never accepts

	fault model	effect	outcome
this sweep	skip the whole bl poly_chknorm	$r_0 = \text{stale} \neq 0 \rightarrow$ always reject	hang
The paper	force the cmp result to 0	$r_0 = 0 \rightarrow$ always accept	leak

4 Plots

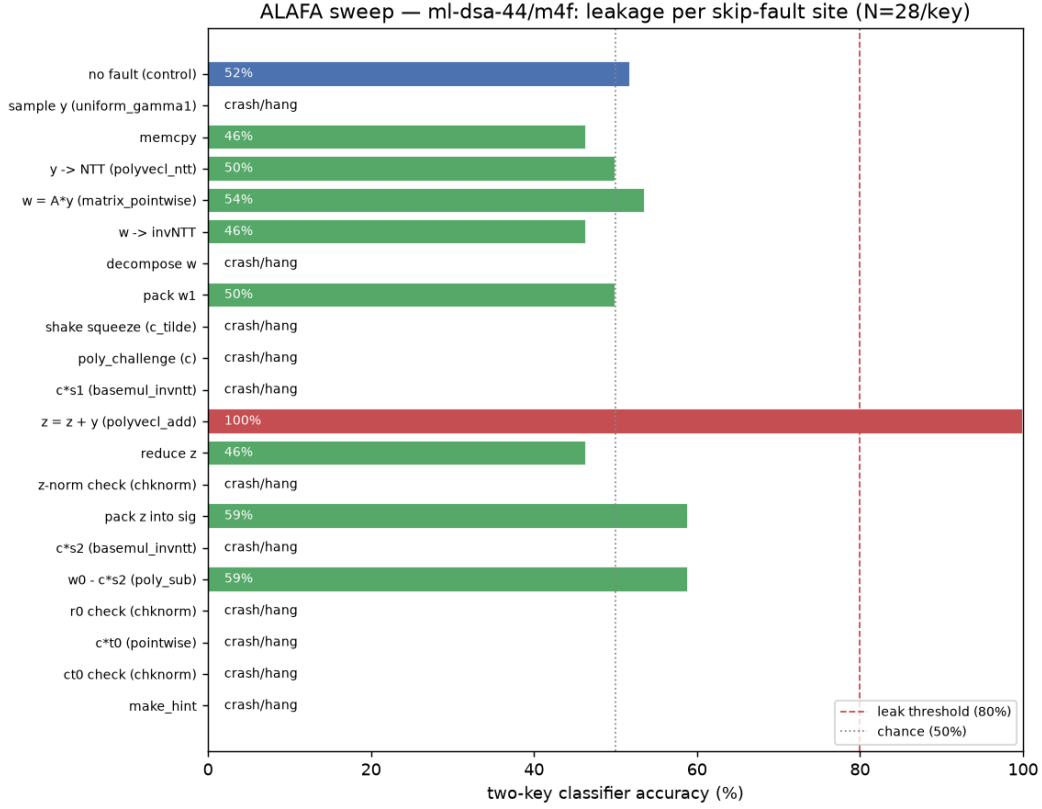


Figure 1: Two-key classifier accuracy per fault site. Blue = control, red = leak, green = ran/no leak, grey = hang. Dashed line = 80% leak threshold, dotted = 50% chance.

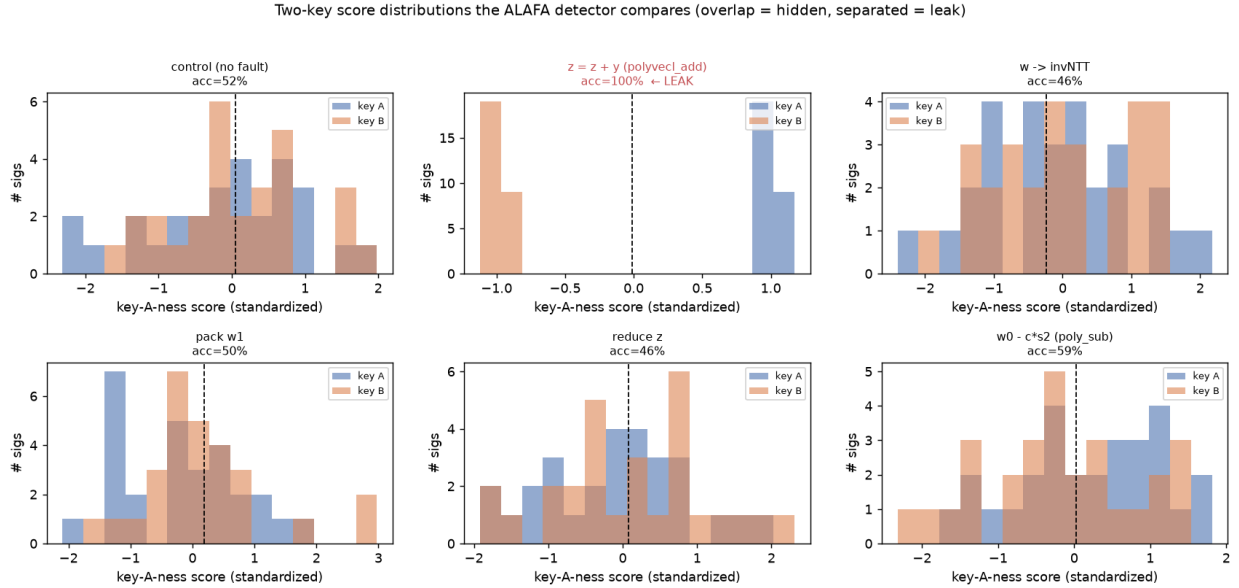


Figure 2: Two-key score distributions at representative sites. Overlap = the output hides the key; separated = leak. Only $z = z + y$ separates.

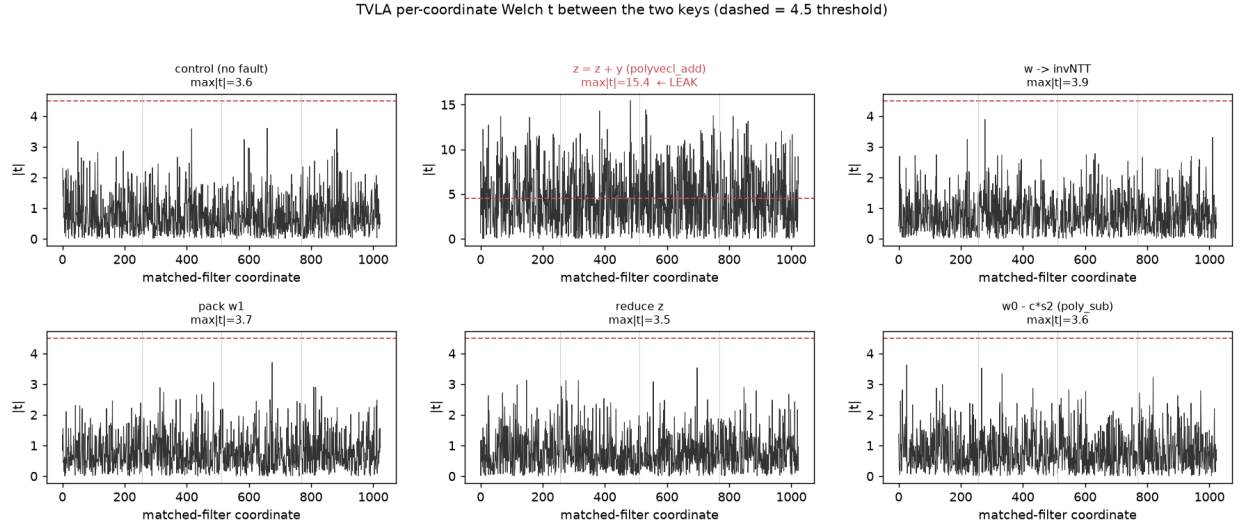


Figure 3: Per-coordinate $|t|$ across the 1024 matched-filter features, one panel per representative site, with the 4.5 threshold dashed. Non-leak sites hug the bottom; only $z = z + y$ is lifted bodily above the line.

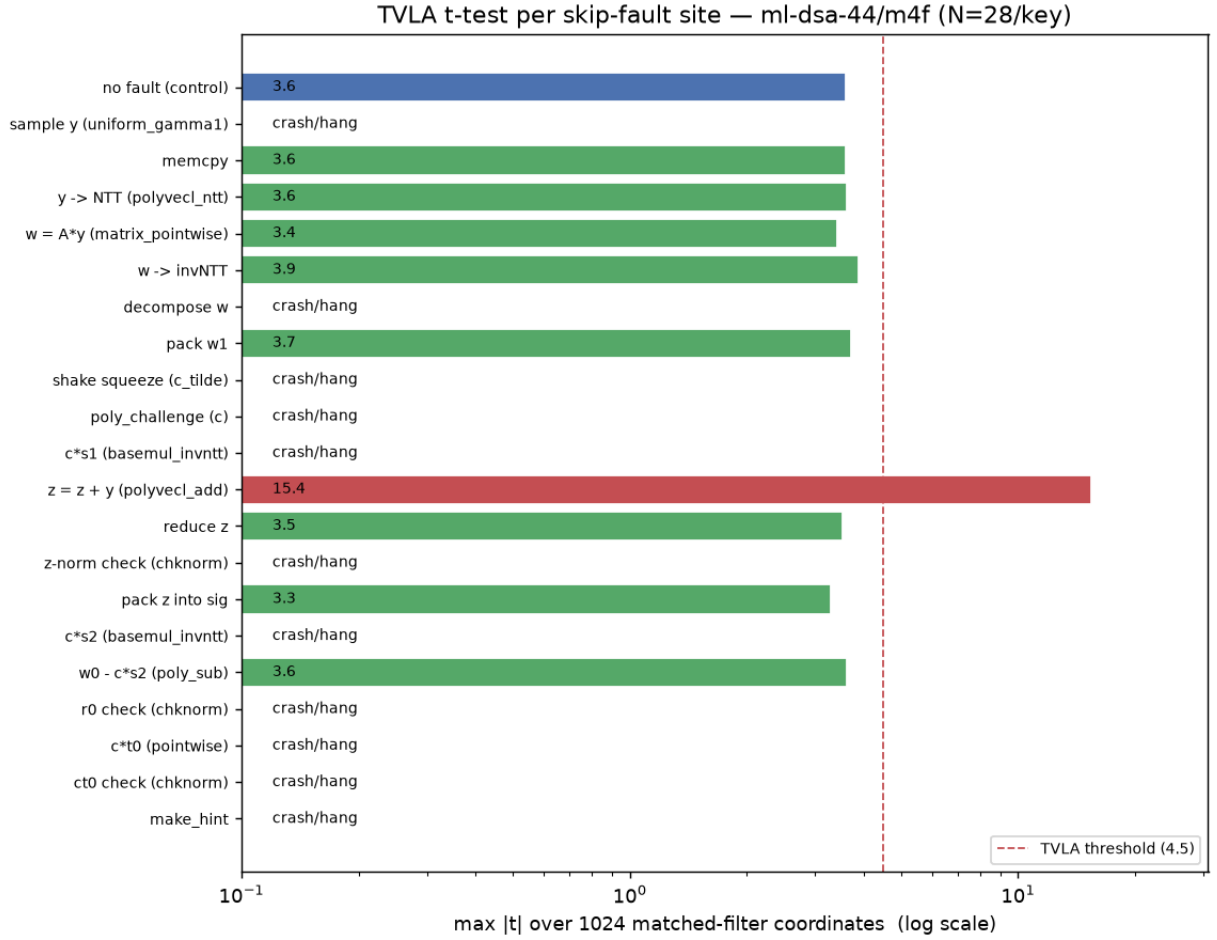


Figure 4: max $|t|$ per fault site (log scale), with the 4.5 threshold